

7.7 Polymerisation (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.7 Polymerisation (A Level Only)
Difficulty	Medium

Time allowed: 50

Score: /40

Percentage: /100

Question 1a

Valylalanine shown in Fig. 1.1, is an example of a dipeptide composed of two amino acids, valine and alanine.

When this compound is placed in acidic conditions, two species are formed.

Draw **one** of these two species.

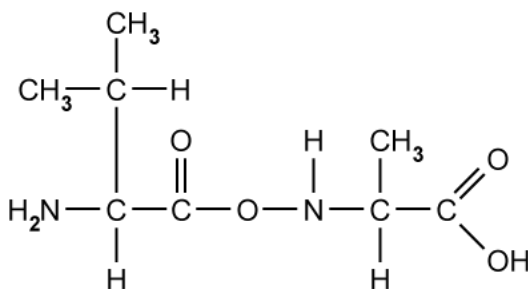


Fig. 1.1

[1 mark]

Question 1b

A dipeptide made from the amino acids cysteine and tryptophan is shown in Fig.1.2. Cysteine contains a sulphur molecule.

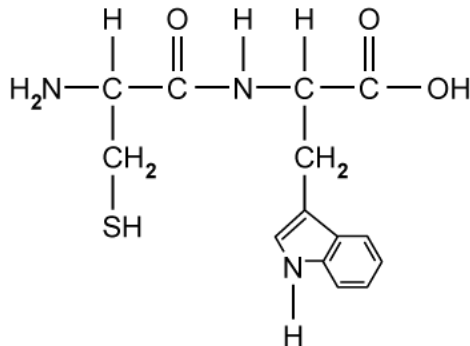


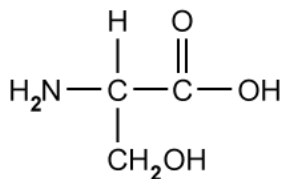
Fig. 1.2

Draw the structure of the amino acid tryptophan.

[1 mark]

Question 1c

Amino acids such as serine shown in Fig. 1.3 will form polyamide links. Polyamides will contain the same type of links and these polymers can be used to form fibres suitable for weaving. Polymers formed from alkenes by additional polymerisation are usually too weak for this purpose.

**Fig. 1.3**

In terms of intermolecular forces between the polymer chains, explain why polyalkenes are not suitable to be used as fibres for weaving.

[3 marks]**Question 1d**

Explain why the molecule $\text{NH}_2\text{CH}_2\text{CH}_2\text{COCl}$ can form a condensation polymer with itself.

[2 marks]

Question 2a

Nylon is an example of a synthetic polyamide and contains the same links as polypeptides. Nylon is the general name for a family of polyamides. A section of a nylon polymer is shown in Fig. 2.1.

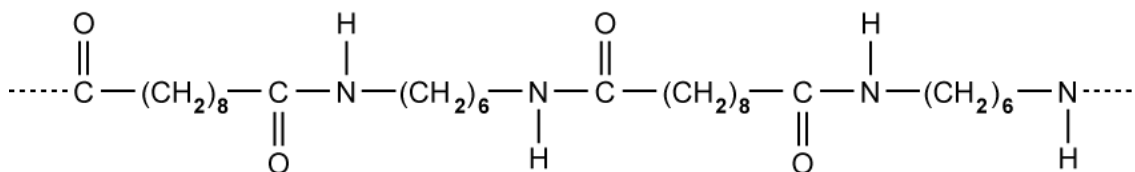


Fig 2.1

i)

Draw the structures of two monomers that could be used to make this nylon.

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[2]

ii)

State the type of polymerisation involved in the formation of this nylon.

[1]

[3 marks]

Question 2b

Nylon can be used to make clothing. Suggest why nylon should be protected from spillages of strong acids and alkalis.

[5 marks]

Question 2c

Nomex shown in Fig. 2.2 has a higher melting point than nylon and is used to make the flame resistant body suits worn by racing drivers.

Suggest a reason why the melting point of Nomex is higher than that of nylon.

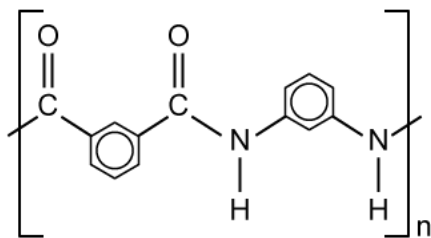


Fig. 2.2

[2 marks]

Question 2d

i)

Draw the **two** monomers that are used to manufacture Nomex

--	--

[2]

ii)

State the formula of any by-products produced.

[1]

[3 marks]

Question 3a

Polyurethanes are polymers made by the reaction of a diisocyanate with a diol as shown in Fig. 3.1. R^1 and R^2 are hydrocarbon groups.

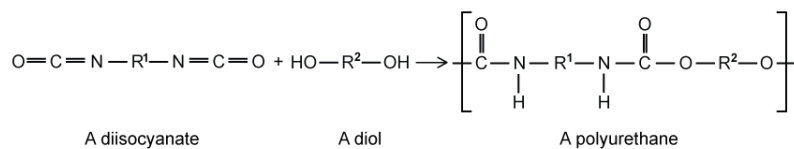


Fig. 3.1

Lycra[®] in Fig. 3.2 is a polyurethane formed from the diisocyanate **P** and $\text{HOCH}_2\text{CH}_2\text{OH}$.

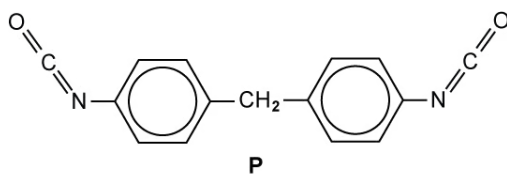


Fig. 3.2

- i)
Give the molecular formula for **P**.

[1]

- ii)
Draw the repeat unit of Lycra[®].

[2]

[3 marks]

Question 3b

Fibres of Lycra[®] are strong due to the intermolecular forces between the polymer chains. Complete the table to identify two intermolecular forces responsible for this property and the group(s) involved.

Intermolecular force	Group(s) involved

[2 marks]**Question 3c**

Name one example of each of the following types of polymer.

Type of polymer	Example
Synthetic polyamide	
Synthetic polyester	

[2 marks]

Question 3d

Fig. 3.3 shows the repeat unit of Nylon 66.

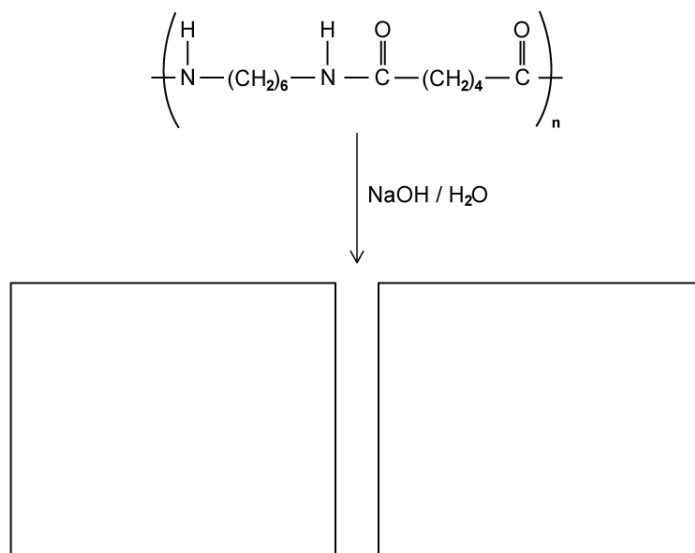


Fig. 3.3

Draw the **two** products from the hydrolysis of Nylon 66.

[2 marks]

Question 4a

i)

Name an example of a synthetic polyester and a synthetic polyamide.

polyester

[1]

polyamide

[1]

ii)

Polyesters and polyamides are formed by condensation reactions.

Name a molecule which is commonly eliminated in such reactions.

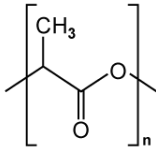
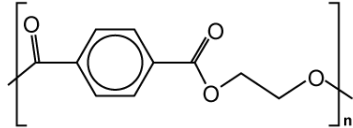
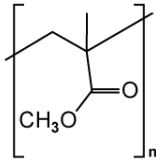
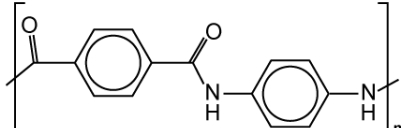
[1]

[3 marks]

Question 4b

i)

The table shows the repeat units of a number of polymers. Place a tick (✓) against the ones which are biodegradable.

Polymer	Repeat Unit	Biodegradable
A		
B		
C		
D		

[2]

ii)

Draw the structures of two monomers used to form polymer B.

[2]

[4 marks]

Question 4c

A section of polypeptide was hydrolysed and the following amino acids were identified.

Amino Acid	Formula
T	$\text{CH}_3\text{CH}(\text{NH}_2)\text{CO}_2\text{H}$
U	$\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{NH}_2)\text{CO}_2\text{H}$
V	$\text{H}_2\text{N}(\text{CH}_2)_4\text{CH}(\text{NH}_2)\text{CO}_2\text{H}$

i)
Which of the amino acids **T**, **U** or **V** has the highest pH in an aqueous solution? Explain your answer.

Amino acid

[1]

ii)
State how many different dipeptides could be formed from a reaction mixture consisting of amino acids **T** and **U**.

[1]

iii)
Polypeptides contain a high proportion of carbon and hydrogen in their structures, yet many are soluble in water. By referring to the structure of a polypeptide, explain why.

[2]

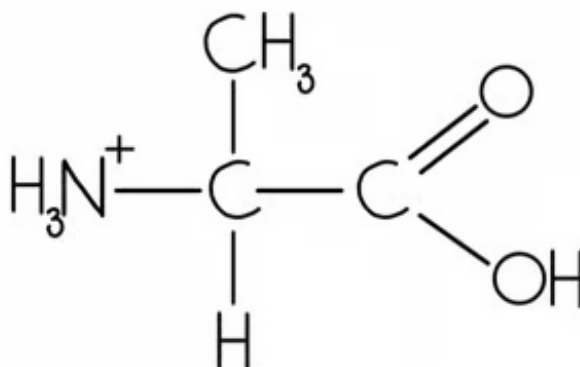
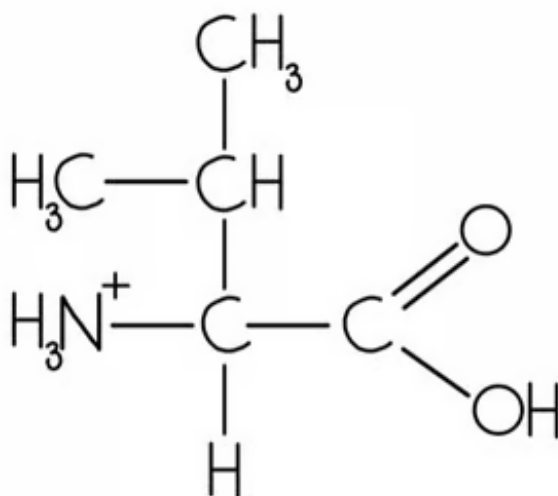
[4 marks]

Model Answers: Medium

1a

a) One species formed from valylalanine in acidic conditions is:

- [1 mark] for either of the correct structures

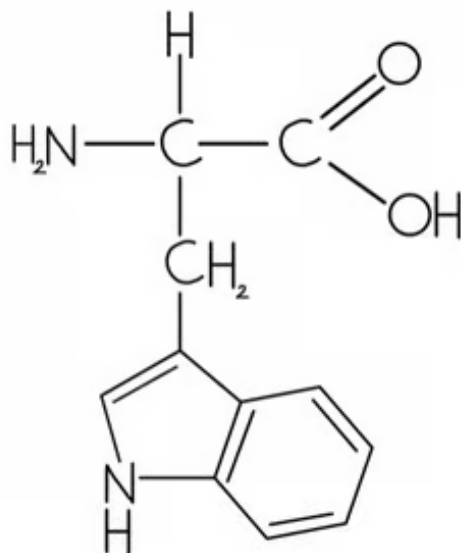


[Total: 1 mark]

- When a dipeptide is placed in acidic conditions a carboxylic acid group will be formed along with a positive nitrogen ion with three hydrogen atoms attached

1b

b) The structure of the amino acid tryptophan is:



[Total: 1 mark]

- Remember to look for the amide link / bond in these structures as they will give the clue as to where the molecule can be separated into the two amino acids
- Look for the C=O bonded to the N-H
- Don't forget to add in the hydrogen atom that will have formed water with the -OH functional group from the COOH in the other amino acid molecule

1c

c) Polyalkenes are not suitable to be used as fibres for weaving because:

- Hydrogen bonds / H bonding present in polyamides

OR

No Hydrogen bonds / H bonding in polyalkenes; [1 mark]

- Only van der Waals forces in polyalkenes; [1 mark]
- Forces in polyalkenes are much weaker than in polyamides

OR

Hydrogen bonds are much stronger than van der Waals forces; [1 mark]

[Total: 3 marks]

- **Remember:** Hydrogen bonding can occur when a hydrogen atom is bonded to nitrogen, oxygen or fluorine
- Hydrogen bonds can not form between strands of addition polymers, but this can occur with condensation polymers

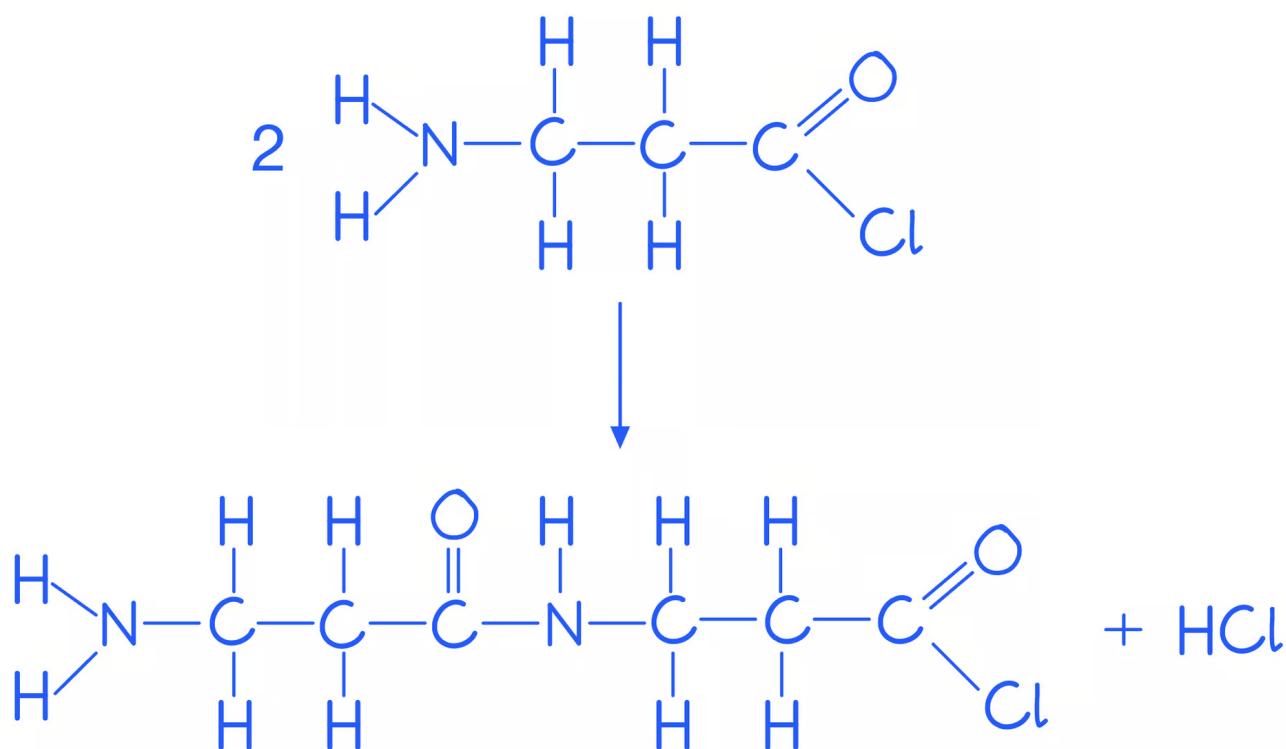
1d

d) The molecule $\text{NH}_2\text{CH}_2\text{CH}_2\text{COCl}$ can form a condensation polymer with itself because:

- $\text{NH}_2\text{CH}_2\text{CH}_2\text{COCl}$ that contains both an amine and acyl chloride group
OR
The molecule contains two different functional groups
OR
The molecule contains functional groups; [1 mark]
- That can / will react with each other; [1 mark]

[Total: 2 marks]

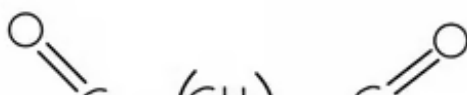
- The compound has the functional groups available to form a condensation polymer
 - The acid chloride group and amino group will react to form a peptide link and eliminate water

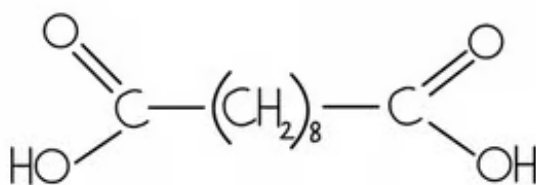


2a

a)

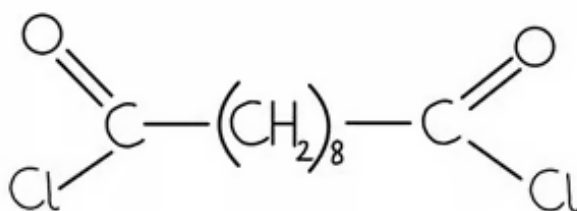
i) The structures of two monomers that could be used to make nylon are:



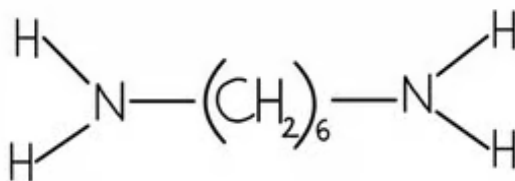


Allow -COOH written as structural formula

OR



; [1 mark]



; [1 mark]

Allow -NH_2 written as structural formula

ii) The type of polymerisation is:

Condensation (polymerisation); [1 mark]

[Total: 3 marks]

- Remember that polyamides/polypeptides are able to be formed from amine groups, NH_2 and with either a carboxylic acids group or an acid chloride group

2b

b) Nylon should be protected from spillages of strong acids and alkalis because:

- Nylon is a polyamide / polypeptide ; [1 mark]
 - Both strong acids and alkalis can hydrolyse polyamide / polypeptide
- OR**
- Spilling strong acids and/or alkalis could cause hydrolysis to occur; [1 mark]
- Polar bond / C=O bond (in the polymer) which is susceptible to attack; [1 mark]
 - Hydrolysis would break the peptide link / bond; [1 mark]
 - Which would create a hole in the fabric; [1 mark]

[Total: 5 marks]

- Hydrolysis is the reaction that polyamides / polypeptides and polyesters can undergo. They react with a strong acid or strong alkali and the link bonds (peptide / amide or ester) are broken
 - This breaks the polymer chain and therefore creates a hole in the fabric
- The presence of the polar C=O bond in polyamides / polypeptides and polyesters is polar which means these polymers can react with acids to form the -COOH group or alkalis to form -COO⁻Na⁺ salts

2c

c) Nomex have a higher melting point than nylon because:

- Nomex has stronger intermolecular forces than nylon; [1 mark]
 - Nomex is planar so can pack together more tightly
- OR**
- Nomex has more surface area points of contact between the molecules
- OR**
- Has greater / stronger hydrogen bonding between molecules
- OR**
- Has greater / stronger van der Waals' forces between molecules; [1 mark]

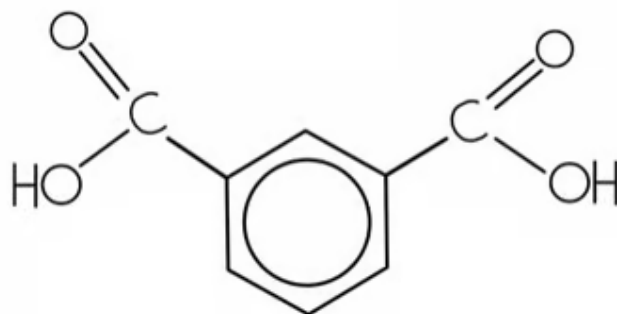
[Total: 2 marks]

- Use both of the images in this question to help you answer the question
- The more intermolecular forces and the more tightly polymer chains can pack together, the higher the melting point
- This is the same as with alkanes and other molecules like you learned in the AS content

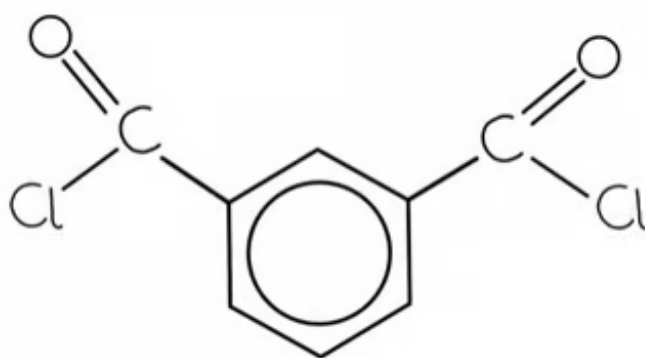
2d

d)

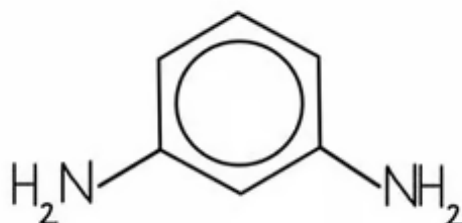
i) The diagram of two monomers used to make Nomex and the by-product formed:



OR



; [1 mark]



; [1 mark]

i) The formula of the by-product(s) are:

- H_2O

OR

HCl; [1 mark]

[Total: 3 marks]

- In order to draw the two monomers the amide/peptide link must be identified and broken
- There must be two amine groups present in one of the monomers and either -COOH or -COCl groups in the other
- The by-product eliminated in the reaction will be H₂O if your monomer contained two -COOH groups, and HCl if your monomer contained two -COCl groups
 - Make sure you give the correct by-product for the monomer you drew
- **Careful:** the question said to give the formula, so you will not give a mark if you give the name instead

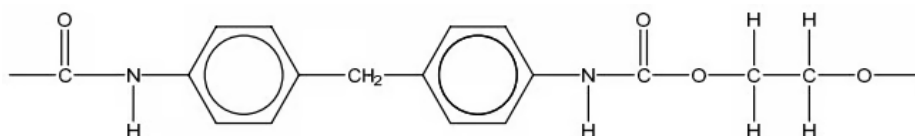
3a

a)

i) The molecular formula for **P** is:

- C₁₅H₁₀N₂O₂; [1 mark]

ii) The repeat unit of Lycra[®] is:



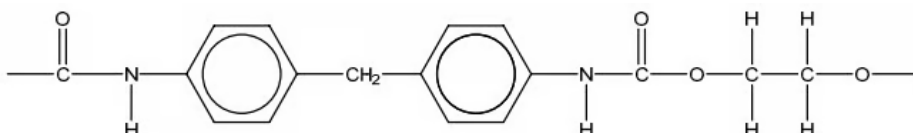
- -NH-CO-O- linkage; [1 mark]
- Rest of the molecule correct; [1 mark]

3a

a)

i) The molecular formula for **P** is:

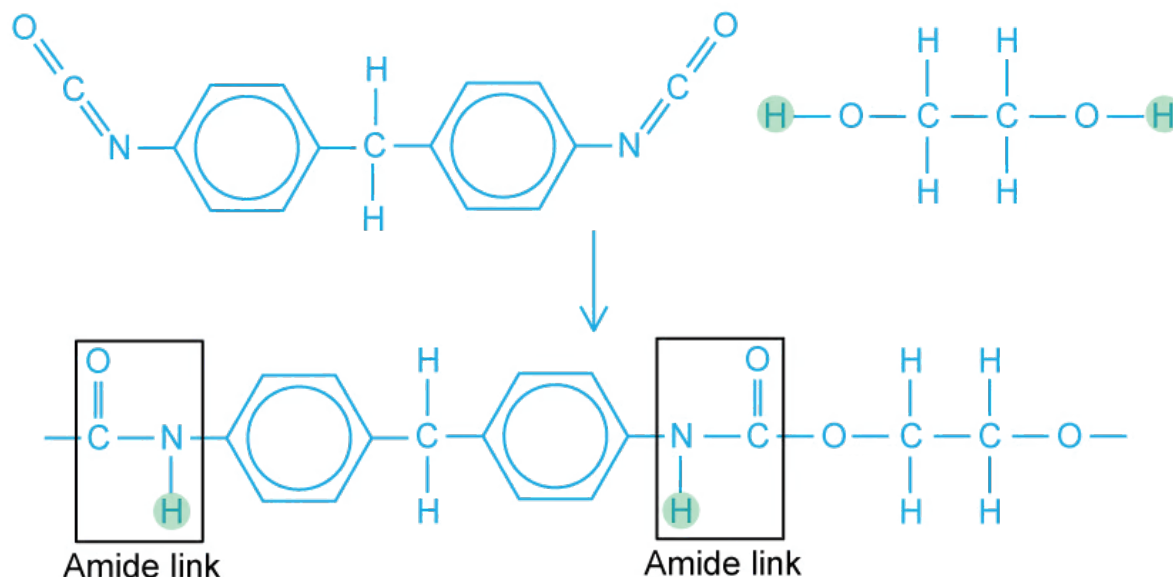
- $C_{15}H_{10}N_2O_2$; [1 mark]

ii) The repeat unit of Lycra[®] is:

- -NH-CO-O- linkage; [1 mark]
- Rest of the molecule correct; [1 mark]

[Total: 3 marks]

- When writing the molecular formula of a compound containing a benzene ring, there will be 'hidden' hydrogens
 - Forgetting to take note of these is a common mistake



- To draw the repeating unit, write the molecules side by side and then look at where the linking bonds can be formed
 - Make sure that either end of the molecule is also correct

- You can use Fig 3.1 as an example to template your answer
 - The R^1 group is the benzene ring- CH_2 group-benzene ring from the diisocyanate molecule **P** in Fig 3.2
 - The R^2 group is $-\text{CH}_2-\text{CH}_2-$ from the specific information given about the diol ($\text{HOCH}_2\text{CH}_2\text{OH}$) in the question
 - You can then replace the R^1 and R^2 groups to get your answer

3b

b) The completed table is:

Intermolecular force	Group(s) involved
hydrogen bonding	NH
VDW forces / Induced dipole-dipole forces / polar forces	$-\text{C}_6\text{H}_4\text{CH}_2-$ allow benzene / aromatic rings

- Hydrogen bonding; [1 mark]
- NH group for hydrogen bonding

AND

Correct second intermolecular force and groups involved ; [1 mark]

[Total: 2 marks]

- Remember:** Hydrogen bonding will occur between molecules that contain a hydrogen atom directly bonded to either an O, N or F atom
- The $-\text{C}_6\text{H}_4\text{CH}_2-$ part of the polymer does not contain any polar bonds therefore the strongest type of intermolecular force must be induced dipole-dipole forces

3c

c) One example of each of the following types of polymer are:

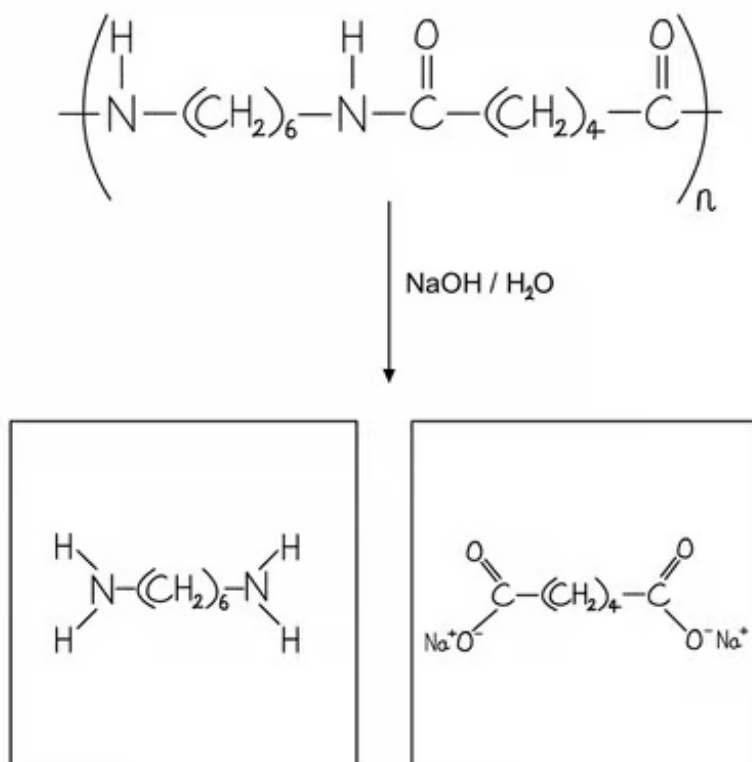
Type of polymer	Example
Synthetic polyamide	Nylon / Kevlar / Nomex; [1 mark]
Synthetic polyester	Terylene / polylactic acid (PLA); [1 mark]

[Total: 2 marks]

- From your studies of polymers at AS and A level, you should be able to give an example of a synthetic polyamide and polyester

3d

d) The two products of hydrolysis are:



- Correct structure for the diamine; [1 mark]
- Correct structure for the sodium salt of the dicarboxylic acid; [1 mark]

[Total: 2 marks]

- Nylon 66 is a polyamide and NaOH / H₂O are the reagents required for alkaline hydrolysis
- The polyamide is heated with sodium hydroxide solution
- Each amide linkage is hydrolysed producing an -NH₂ group and the sodium salt of the acid, ⁻COONa⁺

4a

a)

i) An example of a polyester and polyamide is:

- Polyester : Terylene / polylactic acid (PLA); [1 mark]
- Polyamide : nylon / Kevlar / Nomex; [1 mark]

ii) A molecule which is commonly eliminated in such reactions is:

- Water

OR

Hydrochloric acid / hydrogen chloride; [1 mark]

[Total: 3 marks]

- When a condensation polymer is formed a small molecule is eliminated
 - This does not have to be water which can be a little misleading given the name 'condensation' polymer

4b

b)

i) The polymers which are biodegradable are:

Polymer	Biodegradable
A	✓
B	✓
C	X
D	✓

- 0 - 1 correct; [0 marks]

4b

b)

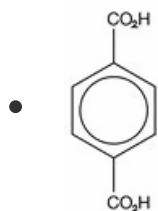
i) The polymers which are biodegradable are:

Polymer	Biodegradable
A	✓
B	✓
C	X
D	✓

- 0 – 1 correct; [0 marks]
- 2 – 3 correct; [1 mark]
- 4 correct; [2 marks]

ii) The structures of the two monomers used to form polymer **B**:

- $\text{HOCH}_2\text{CH}_2\text{OH}$; [1 mark]

**OR**

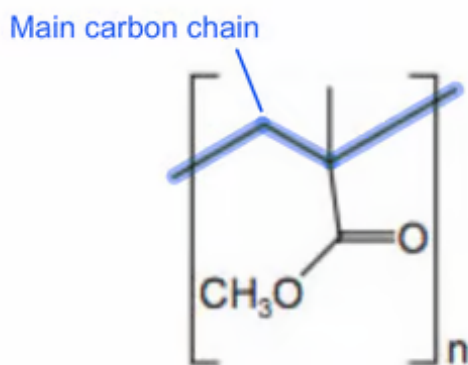
Equivalent 1,4–diacyl chloride

OR

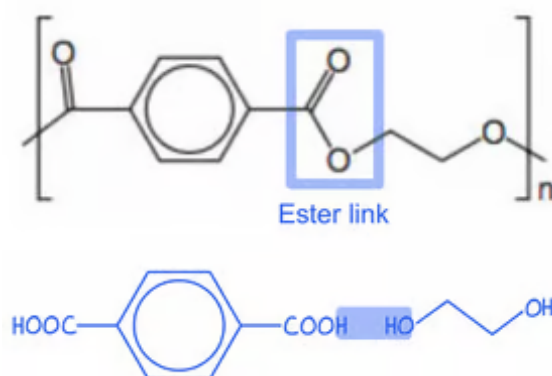
Equivalent 1,4–diester

[Total: 4 marks]

- Polymers **A**, **B** and **D** are condensation polymers and are therefore biodegradable
- Polymer **C** is an addition polymer and therefore is very un-reactive



- To identify the monomers that are used to form polymer **B**:
 - Identify the type of link that forms the polymer
 - If you assume that the molecule eliminated is water then you can draw the dicarboxylic acid and diol



Remember: Ester links are commonly formed from the reaction of an alcohol and a carboxylic acid

- The monomer could be a diacyl chloride rather than a dicarboxylic acid
 - This would mean that the molecule eliminated would be HCl

4c

c)

i) The amino acid that has the highest pH in an aqueous solution is:

- Amino acid: **V**

AND

It has two amine / NH_2 groups (which can be protonated) **OR** it has an amine / NH_2 group on its side chain / R group; [1 mark]

ii) The number of different dipeptides that could be formed from a reaction mixture

consisting of amino acids **T** and **U** is:

- 4 / Four; [1 mark]

iii) Polypeptides contain a high proportion of carbon and hydrogen because:

- Hydrogen bonds; [1 mark]
- Between the O / N atoms or named group (in the polypeptide) and water; [1 mark]

OR

- Ion-dipole attractions; [1 mark]
- Between NH_3^+ / CO_2^- and water; [1 mark]

[Total: 4 marks]

- Amino acid **V** has the highest pH or is the most basic because there are more NH_2 groups present compared to **T** and **U**
- A Brønsted-Lowry base is a proton acceptor
 - If there are more NH_2 groups then the amino acid has a greater capacity to accept protons
- A dipeptide is made up of two amino acids
 - The amino acid can be the same in a dipeptide
 - The amino acids can bond from different sides
- Therefore, there are four possibilities if there are two different amino acids
 - TT, TU, UT, UU
- Hydrogen bonding can occur between molecules where an H atom is bonded to an O, N or F atom
 - Molecules that exhibit hydrogen bonding can interact with water and will be soluble
- An internal acid-base reaction can take place, whereby a proton is transferred from the acid $-\text{COOH}$ group to the basic $-\text{NH}_2$ group to form a bipolar ion called a **zwitterion**